

C - PROGRAMMING

Course Duration: 20 hrs

S. No	Content
1	INTRODUCTION 1. Introduction to C Programming 2. History and Overview of C 3. Importance of C in software development 4. Applications of C 5. Basic Structure of a C Program 6. Example Program
2	Variable • Naming rules • Variables declaration Datatypes • Primitive Datatype • Non-Primitive Datatypes
3	Constants • What is constants • Creating constant variable... • Using const keyword • Using #Define keyword Literals

	Storage Classes • Introduction to storage classes. • Types of Storage Classes.
4	Operators 1. Arithmetic 2. Assignment 3. Relational 4. Misc 5. Logical 6. Bitwise 7. Conditional
5	Decision Making (Branching) 1. If Statement 2. If – Else 3. Nested If 4. If else if Ladder 5. Switch case
6	Flow Control 1. For Loop 2. While Loop 3. Do – While Loop

7	Jump Statement 1. Break 2. Continue 3. goto
8	Functions • Introduction to functions. • Advantage of Functions • Types of Functions • Recursion
9	Array • Initialization of an array • Accessing array elements • Types of an array 1. Single Dimensional Array 2. Multi-Dimensional Array a) Two-Dimensional b) Three-Dimensional • Sample program using array
10	String • Introduction to Strings • Array of String • Standard String Library Functions • Sample Program

11	Pointer • Introduction to Pointers • Pointers attribute • Pointers as function parameter • Pointers and Array • Pointers to pointers
12	Structure • Introduction to Structure • Using struct keyword • Sample Program Union • Introduction to Union • Difference between structures and unions • Using union keyword • Sample Program
13	Enum • Introduction to Enum • Using enum keyword • Sample Program TypeDef • Introduction to TypeDef • Using typedef keyword • Sample Program

14	Dynamic Memory • Introduction to Dynamic Memory allocation • Allocation using Malloc() • Allocation using Calloc() • Allocation using Realloc() • Allocation using Free()
15	Error Handling • Introduction to Errors • Using ERRNO Variable • ERRNO ValueError • Sample program
16	File I/O Handling • Introduction to Input/Output file Handling • Creating a file in desktop Using Functions like: • Fopen to open the file • Fclose to close the file • Fscanf to read the file • Fprintf to write in a file • Read & write using fput() and fget() functions. • Sample programs
17	Exercise & Practice